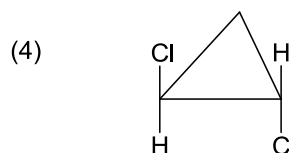
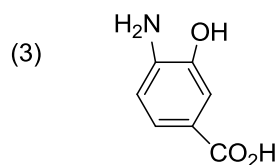
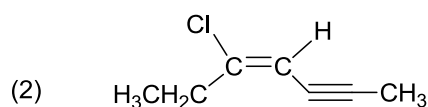
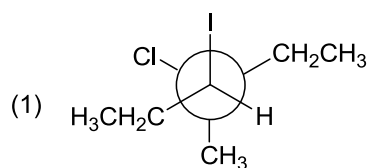


一	二	三	四	五	总分

请直接在试卷上作答

一、 回答下列问题：29 分

1、用系统命名法命名下列化合物(8 分)



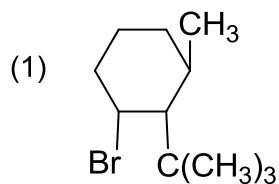
(1)

(2)

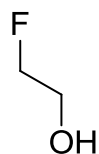
(3)

(4)

2. 写出下列化合物最稳定的构象式(2 分)

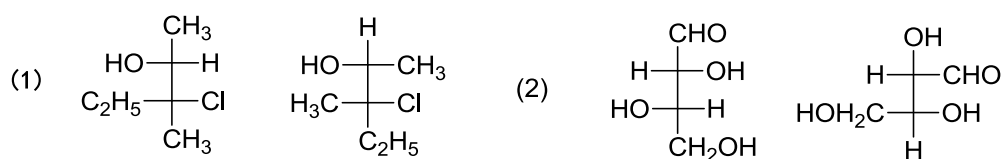


(2)

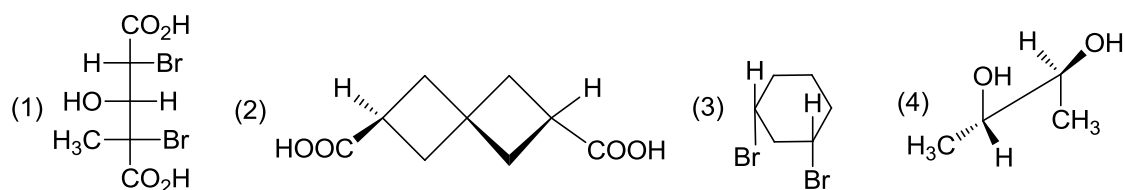


(请画出纽曼式)

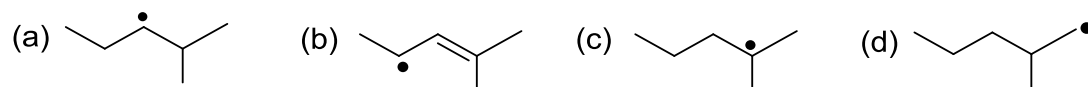
3. 判断下列各对投影式之间的关系 (4 分)



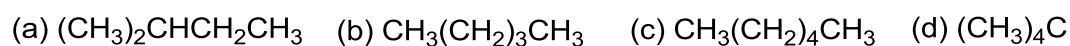
4. 判断下列化合物是否有手性 (4 分)



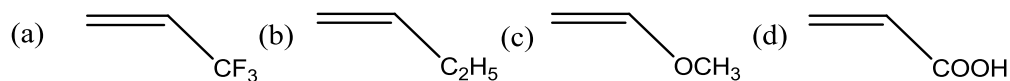
5. 比较下列碳自由基的稳定性大小 (由大到小排列) (2 分)



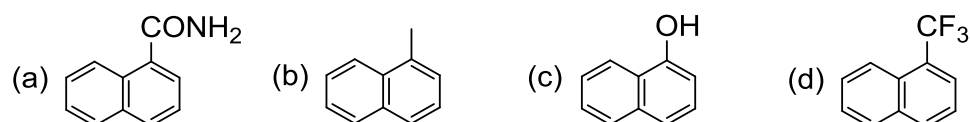
6. 比较下列化合物的沸点顺序 (由大到小排列) (1 分)



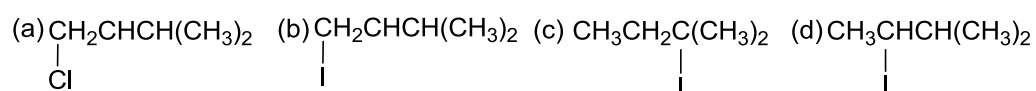
7. 比较下列化合物发生亲电加成反应的活性顺序 (由大到小排列) (1 分)



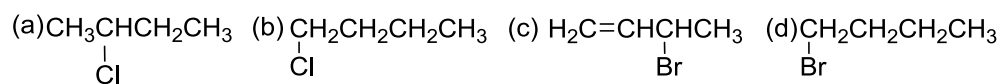
8. 比较下列化合物发生芳香亲电取代反应的活性顺序 (由大到小排列) (1 分)



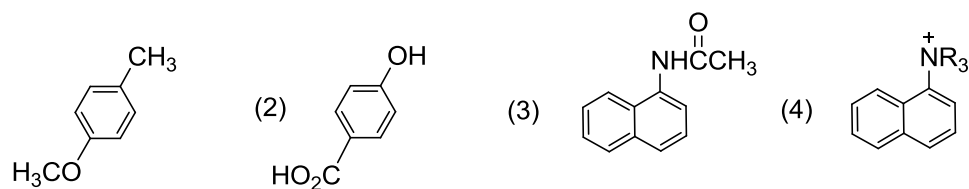
9. 比较下列化合物发生 S_N1 反应的活性顺序 (由大到小排列) (1 分)



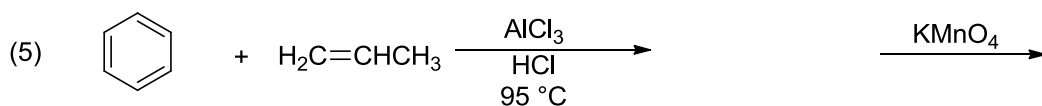
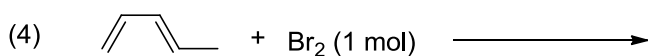
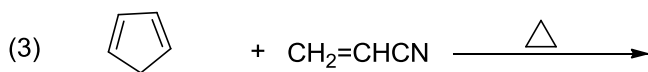
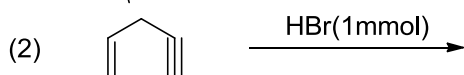
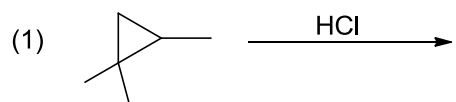
10. 比较下列化合物发生 S_N2 反应的活性顺序 (由大到小排列) (1 分)

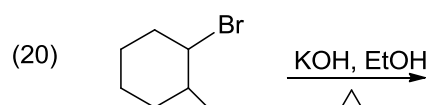
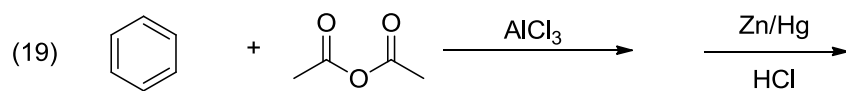
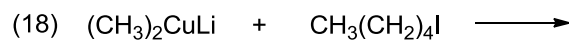
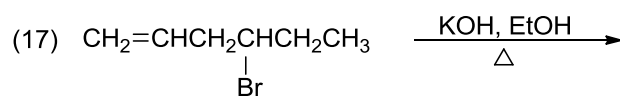
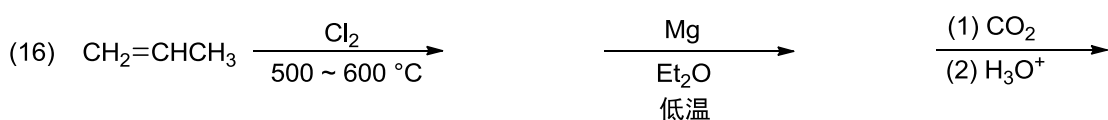
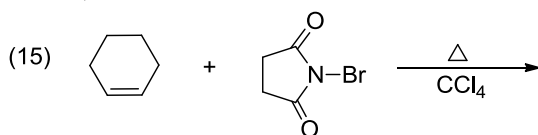
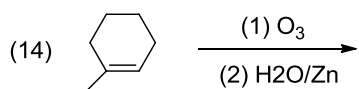
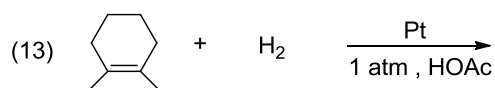
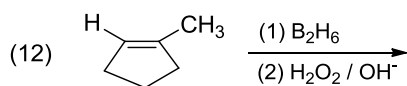
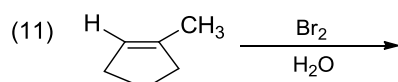
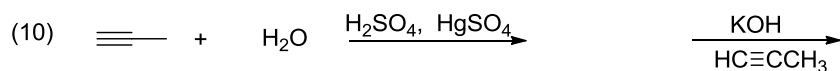
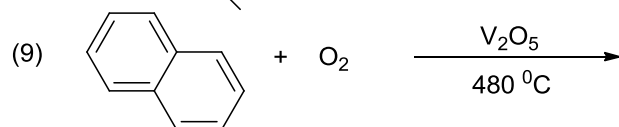
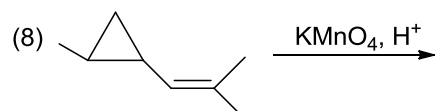
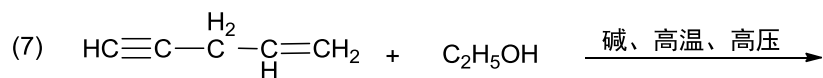
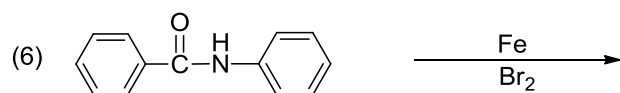


11. 用箭头表示下列化合物发生硝化反应时, 取代基主要进入苯环的位置 (4 分)



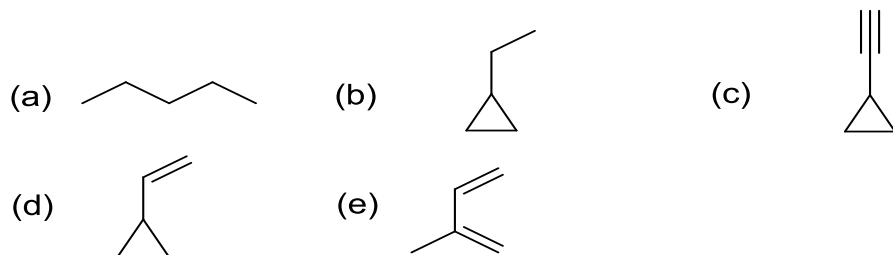
二、 写出下列反应的主要产物 (必要时注明立体化学问题) (25 分)



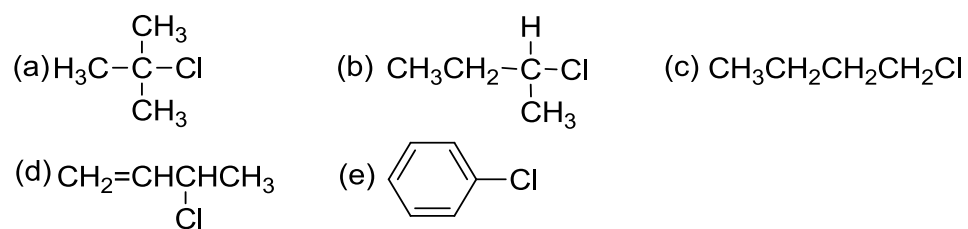


三、 鉴别题（10 分）

（1）用简单的化学方法鉴别下列各组化合物（5 分）

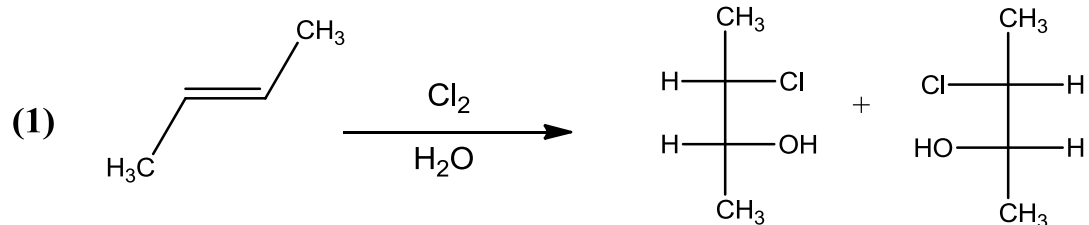


（2）用简单的化学方法鉴别下列卤化物（5 分）

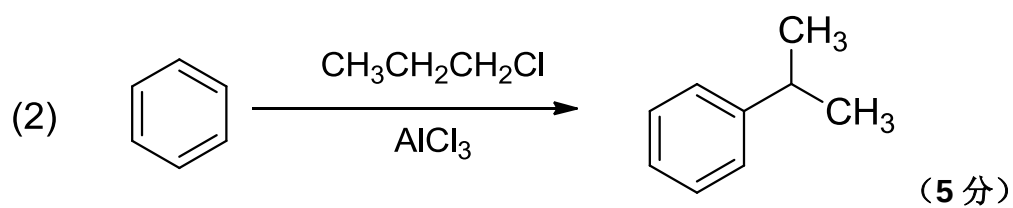


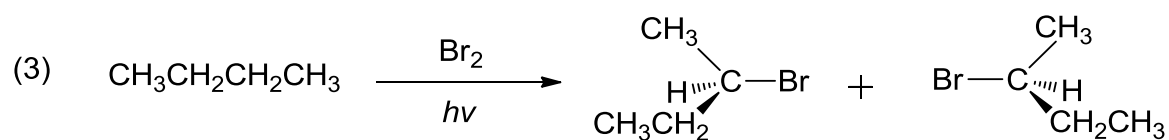
四、 反应历程推测题（18 分）

请写出下列反应的机理（用箭头表示电子对的转移，用鱼钩箭头表示单电子的转移，并画出反应可能存在的中间体）。

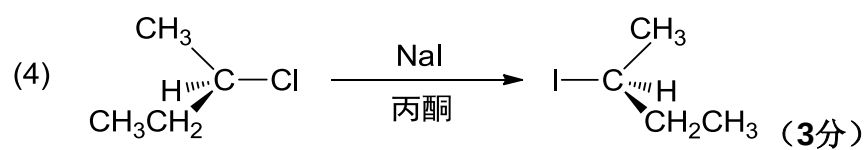


（需标明两种产物各自的反应途径）（5 分）





(5 分)



五、合成题：(18 分)

请用苯、甲苯及含碳原子数小于两个（包括两个）的有机化合物为起始原料合成下列化合物(无机试剂任选)。

